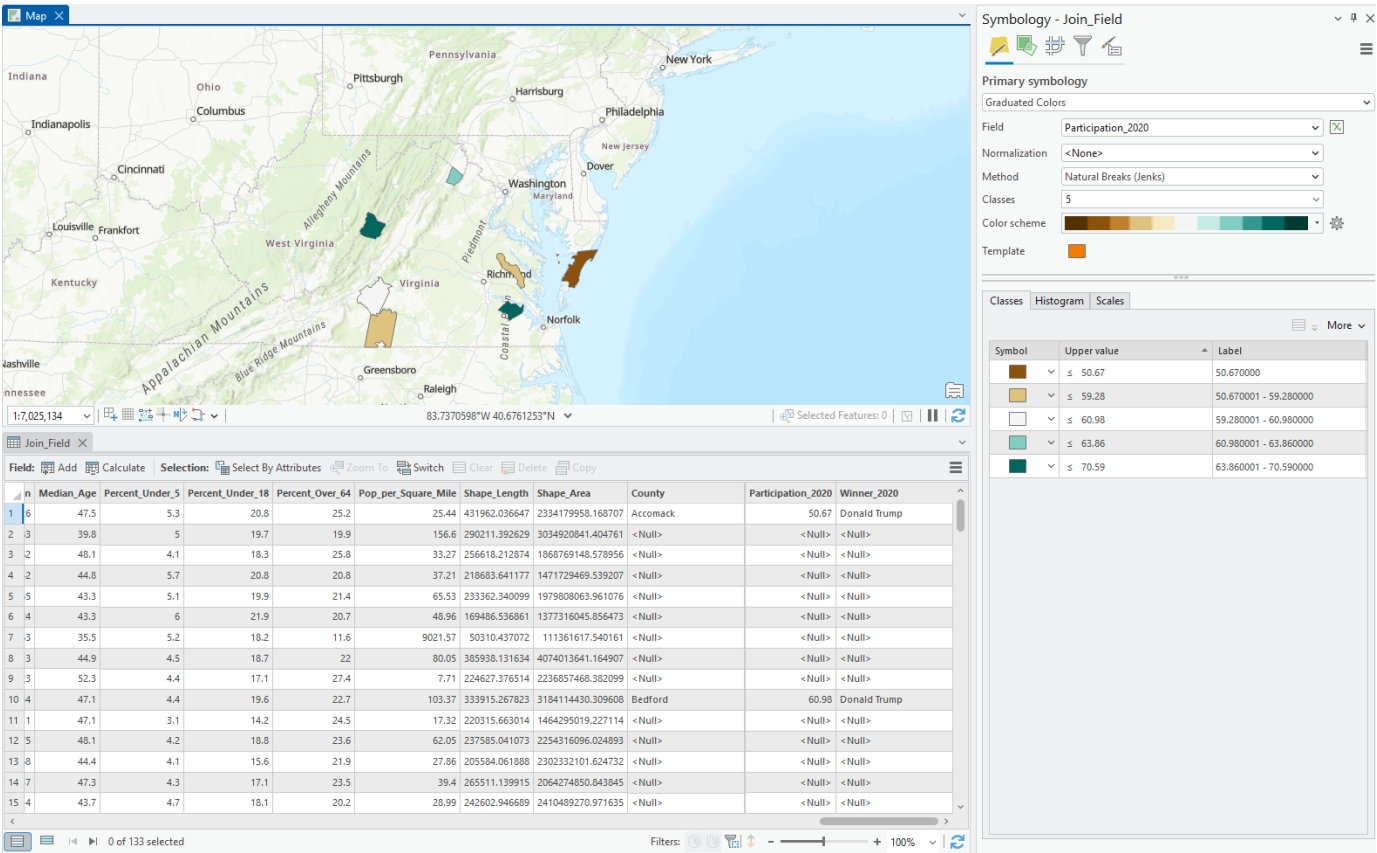


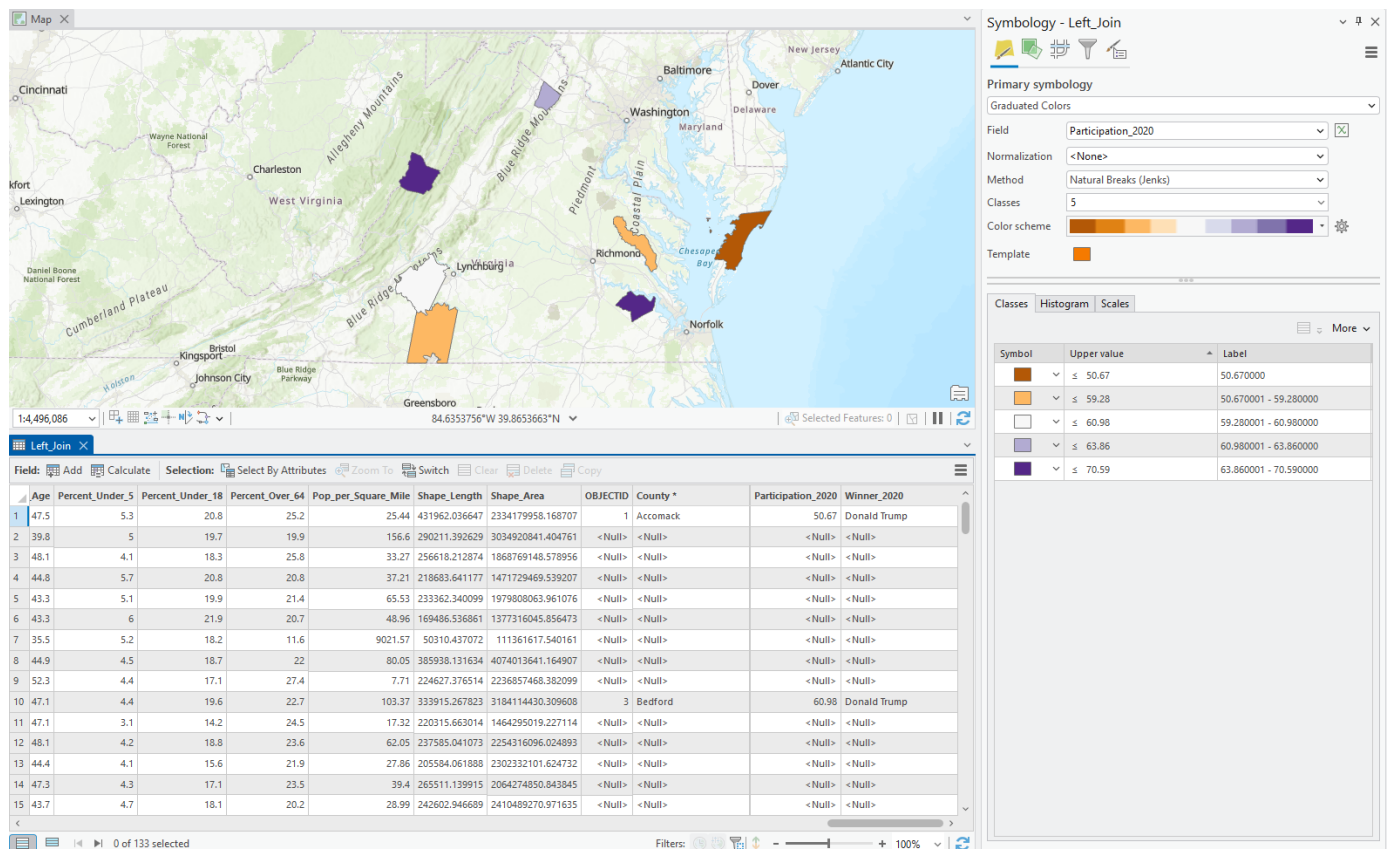
Week 8 - Joins Revision

Henry Rotunno

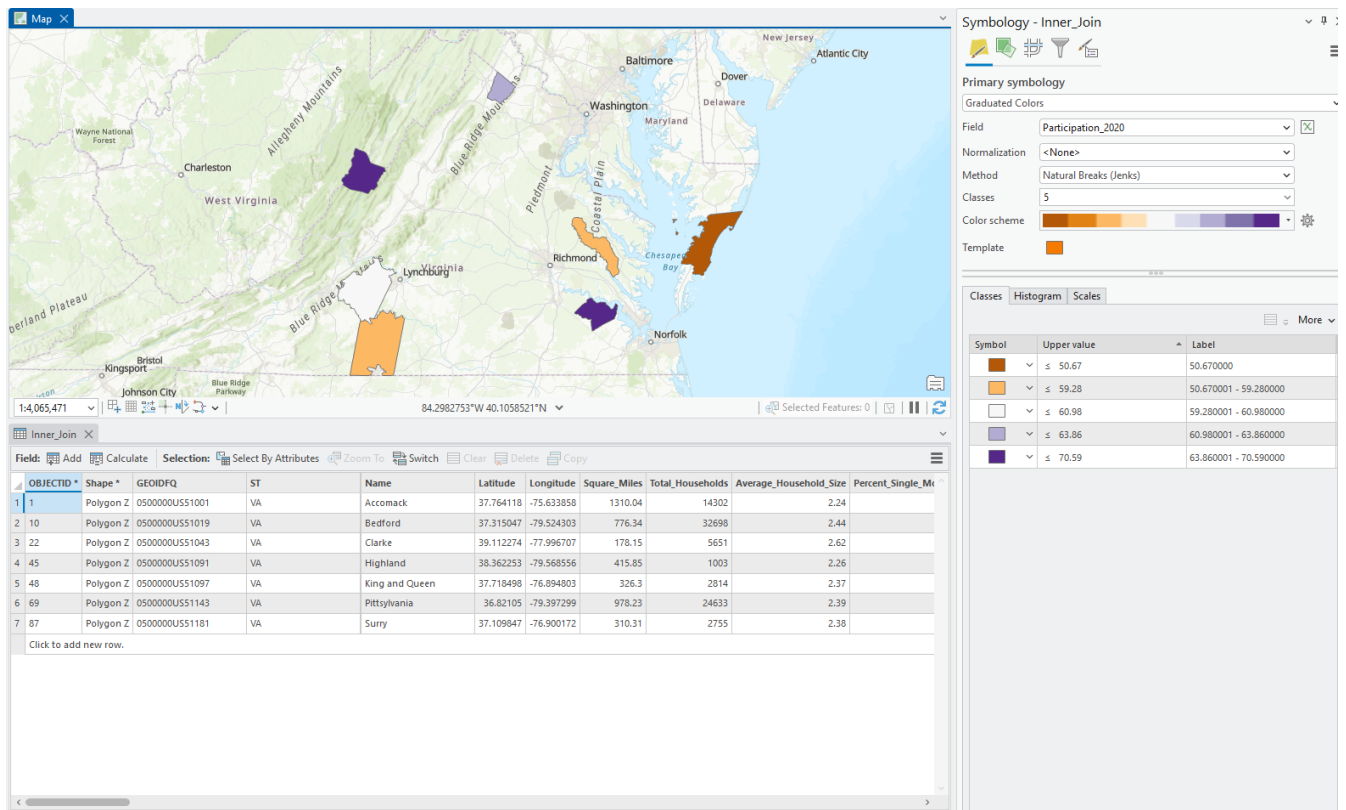
24 October 2025



Join field



Left join



Inner join

SQL command

```
SELECT counties.*, participation.participation_2020
FROM counties
INNER JOIN participation
ON counties.name = participation.county
```

objectid	geoidfq	st	name	latitude	longitude	square_miles	total_households	average_household_size	percent_single_mothers	percent_in_college	percent_bachelors_degree	percent
1	05000000U51001	VA	Accomack	37.76411782	-75.63385825	1310.040000000	14302.000000000	2.240000000	54.600000000	17.000000000	12.700000000	
10	05000000U51019	VA	Bedford	37.31504738	-79.52430318	776.340000000	32698.000000000	2.440000000	18.400000000	23.600000000	20.800000000	
22	05000000U51043	VA	Clarke	39.11227449	-77.99670676	178.150000000	5651.000000000	2.620000000	33.800000000	21.600000000	23.200000000	
45	05000000U51091	VA	Highland	38.36225305	-79.56855643	415.850000000	1003.000000000	2.260000000	87.300000000	20.400000000	22.200000000	
48	05000000U51097	VA	King and Queen	37.71849771	-76.89480274	326.300000000	2814.000000000	2.370000000	57.400000000	13.000000000	14.300000000	
69	05000000U51143	VA	Pittsylvania	36.82105006	-79.39729916	978.230000000	24633.000000000	2.390000000	39.200000000	21.900000000	10.700000000	
87	05000000U51181	VA	Surry	37.10984702	-76.90017178	310.310000000	2755.000000000	2.380000000	41.400000000	30.300000000	18.300000000	
102	05000000U51580	VA	Covington	37.77852350	-79.98676653	5.670000000	2493.000000000	2.240000000	35.700000000	25.500000000	6.000000000	

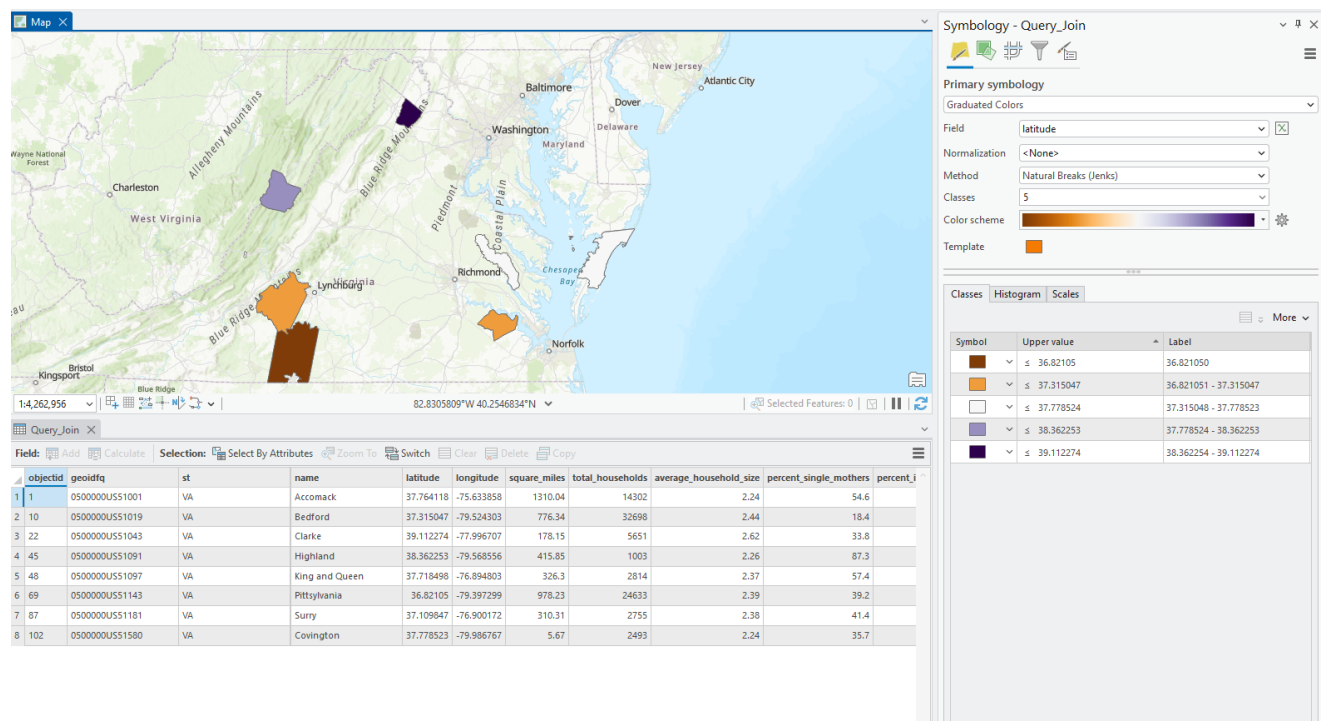
8 rows (0.034 s) Edit, Explain, Export

```
SELECT counties.*, participation.participation_2020
FROM counties
INNER JOIN participation
ON counties.name = participation.county
```

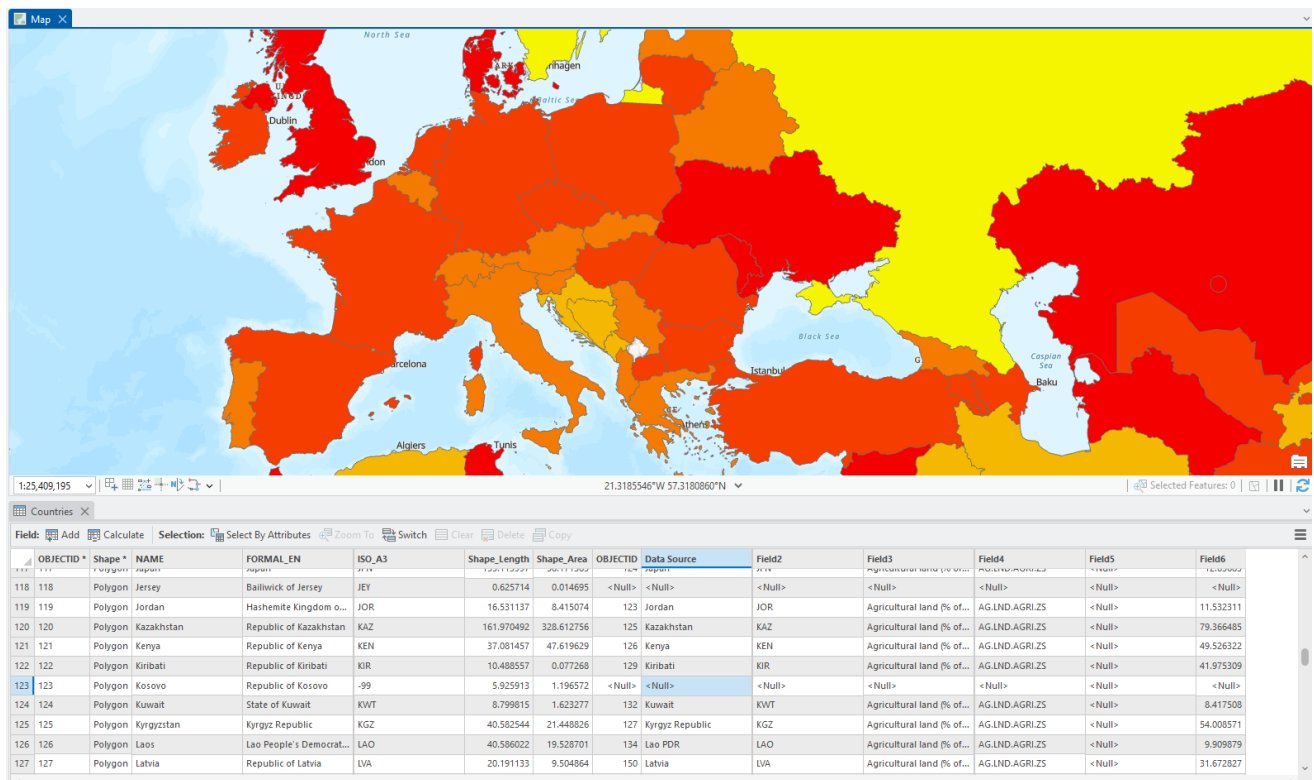
Execute Limit rows: ☐ Stop on error ☐ Show only errors

History

Console inner join



Query layer join



Mismatched names in a join

magery

Share

Help

Table

Standalone Table

Versioning

Catalog Pane

Geoprocessing

Workflow Manager

Create

Add

Import

Remove

Map

Contents

Validate Join

Start Time: Thursday, October 16, 2025 3:36:33 PM

Checking for invalid characters...

Checking workspaces...

Checking for field indexes...

The join field ZCTA5CE20 in the table countydata.Zip_Codes is not indexed. To improve performance, we recommend that an index be created.

The join field ZIP_Code in the table countydata.Cancer_Table is not indexed. To improve performance, we recommend that an index be created.

Checking for OIDs...

Checking for join cardinality (1:1 or 1:m joins)...

WARNING 003237: Join had no matches.

The input table has 33791 and the join table has 1226 records.

Succeeded at Thursday, October 16, 2025 3:36:44 PM (Elapsed Time: 10.49 seconds)

Close

Add Join

Input Table

Zip_Codes

Input Field

ZCTA5CE20

Join Table

Cancer_Table

Join Field

ZIP Code

Keep all input records

☒

Index join fields

☐

Join Operation

Validate Join

OK

1:16,261,885

87.5063252°W 39.1382968°N

Selected Features: 0

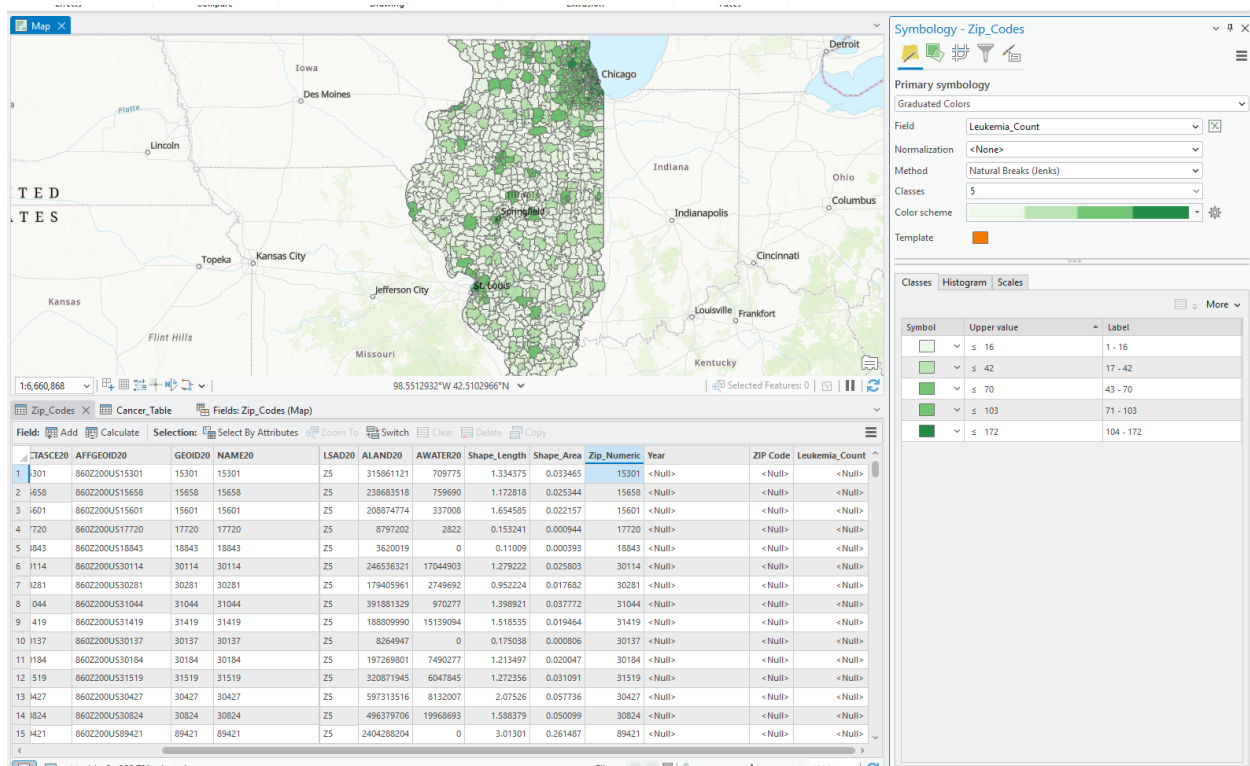
Zip_Codes

Cancer_Table

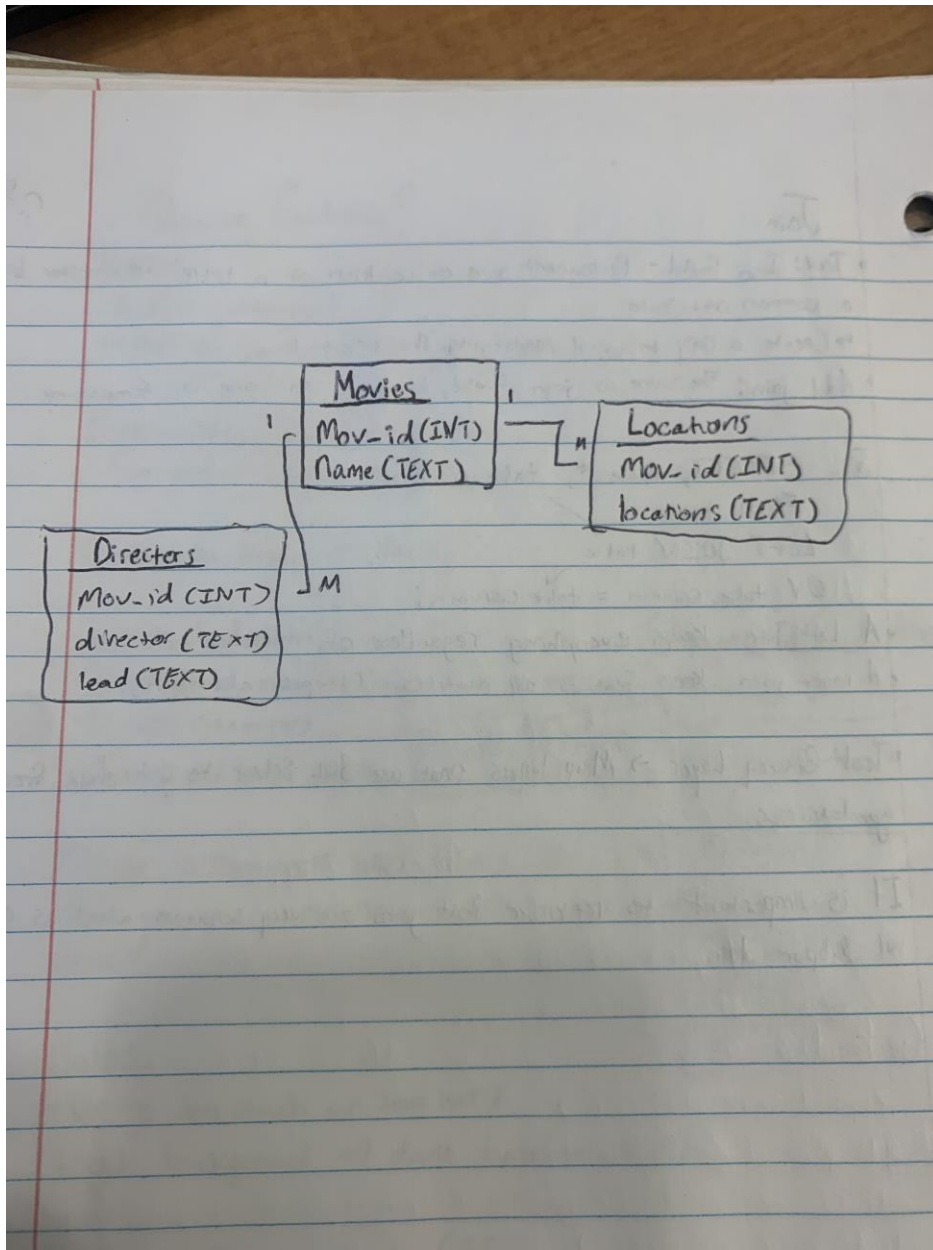
field: Add Calculate Selection: Select By Attributes Zoom To Switch Clear Delete Copy

OBJECTID *	Shape *	ZCTA5CE20	AFFGEOID20	GEOID20	NAME20	LSAD20	ALAND20	AWATER20	Shape_Length	Shape_Area
1	Polygon	15301	8602200U15301	15301	15301	Z5	315861121	709775	1.334375	0.033465
2	Polygon	15658	8602200U15658	15658	15658	Z5	238683518	759690	1.172818	0.025344
3	Polygon	15601	8602200U15601	15601	15601	Z5	208874774	337008	1.654585	0.022157
4	Polygon	17720	8602200U17720	17720	17720	Z5	8797202	2822	0.153241	0.000944
5	Polygon	18843	8602200U18843	18843	18843	Z5	3620019	0	0.11009	0.000393
6	Polygon	30114	8602200U30114	30114	30114	Z5	246536321	17044903	1.279222	0.025803
7	Polygon	30281	8602200U30281	30281	30281	Z5	179405961	2749692	0.952224	0.017682
8	Polygon	31044	8602200U31044	31044	31044	Z5	391881329	970277	1.398921	0.037772
9	Polygon	31419	8602200U31419	31419	31419	Z5	188809990	15139094	1.518535	0.019464
10	Polygon	30137	8602200U30137	30137	30137	Z5	8264947	0	0.175038	0.000806
11	Polygon	30184	8602200U30184	30184	30184	Z5	197269801	7490277	1.213497	0.020047
12	Polygon	31519	8602200U31519	31519	31519	Z5	320871945	6047845	1.272356	0.031091
13	Polygon	30427	8602200U30427	30427	30427	Z5	597313516	8132007	2.07526	0.057736
14	Polygon	30824	8602200U30824	30824	30824	Z5	496379706	19968693	1.588379	0.050099
15	Polygon	89421	8602200U89421	89421	89421	Z5	2404288204	0	3.01301	0.261487

Mismatched key data types problem



Successful join of mismatched key data types



E-R diagram of movies

Select: movies

[Select data](#)
[Show structure](#)
[Alter table](#)
[New item](#)

`SELECT * FROM "movies" LIMIT 50` (0.002 s) [Edit](#)

☐ Modify	objectid	mov_id	name
☐ edit	1	1	All Quiet on the Western Front (1930)
☐ edit	2	2	Halloween
☐ edit	3	3	Oppenheimer

[Import](#)

E-R table

objectid	mov_id	locations
1	1	California
2	2	California

2 rows (0.002 s) [Edit](#), [Explain](#), [Export](#)

```
SELECT * FROM locations WHERE locations.locations = 'California'
```

E-R select

```
SELECT movies.*, director.director, director.lead
FROM movies
LEFT JOIN director
ON movies.mov_id = director.mov_id
```

objectid	mov_id	name	director	lead
1	1	All Quiet on the Western Front (1930)	Lewis Milestone	Lew Ayres
2	2	Halloween	John Carpenter	Jamie Lee Curtis
3	3	Oppenheimer	Christian Bale	Cilian Murphy

3 rows (0.004 s) Edit, Explain, Export

E-R Join

```
SELECT COUNT(locations) FROM locations GROUP BY (locations.locations)
```

count
1
1
1
1
1

5 rows (0.002 s) Edit, Explain, Export

```
SELECT COUNT(locations) FROM locations GROUP BY (locations.locations)
```

E-R aggregation